Attack-as-Design: Turning Surrogate Exploitation into Validated Biological Sequence Design

Ajay Rangarajan Jeyashree Krishnan

Abstract

The same search that produces *adversarial examples*—driving a model's prediction up by making small changes to its input—can be turned into a tool for designing biological sequences. We train a small model to predict a property (the *surrogate*) and then run an attack-style search to find sequences the surrogate scores highly. On its own this search cheats: it finds sequences that fool the surrogate into predicting a high score rather than sequences that are genuinely good. We make the search useful by adding two things. First, a simple biological rule that keeps every proposed sequence realistic, using no extra trained model: each change from the starting sequence must be conservative under the BLOSUM62 (BLOcks SUbstitution Matrix) table, and a set of whole-sequence biophysical numbers (hydrophobicity, net charge, aromatic and charged fractions, sequence complexity, and the longest repeated-letter run) must stay within the ranges seen in natural sequences. Second, and most important, we check every design against signals the search never used—so the final claim never rests on the surrogate grading its own work. We demonstrate the system on **four tasks across two molecule types**: protein thermostability and green fluorescent protein (GFP) fluorescence, a 600-base-pair regulatory-DNA task verified on a true functional model (Borzoi), and a small DNA benchmark with an exact ground-truth answer (TF-Bind-8) used as a diagnostic. The biological rule keeps 100% of GFP designs realistic versus 5–19% for the baselines (paired Wilcoxon $p < 10^{-7}$); on thermostability it matches a learned second model and beats a hand-tuned penalty ($p < 0.001$); and on regulatory DNA the system raises the true property on all 40 of 40 starting points. Independent checks (a structure predictor, a separate thermostability predictor, kept functional residues, and a true DNA-activity model) agree the designs are real. Which featurization the surrogate uses (a frozen foundation-model embedding or a plain one-hot/k-mer encoding) is a swappable part of the system, reported as an ablation, not the point: the foundation model only helps on long, variable-length protein, and the system works either way.

1 Introduction

Many sequence design problems are posed as optimizing a measured property y over sequences x: find the DNA, RNA, or protein sequence that maximizes binding affinity, thermostability, or activity. Because direct measurement is costly, a common approach is to train a *surrogate* model that predicts y from x using available measurements, and then search for sequences the surrogate scores highly. This is the setting of offline model-based optimization [Trabucco et al., 2021, 2022].

This approach has a well-known weakness. A surrogate is only accurate near the sequences it was trained on. When the search pushes into unusual sequences, the surrogate can assign them a high score by mistake, and the search happily returns those mistakes instead of genuinely good sequences. This is exactly the mechanism behind an "adversarial example" in image classification, where a small, deliberately chosen change fools a model into a confident wrong answer. The offline optimization literature already makes this connection [Trabucco et al., 2021, 2022, Fu and Levine, 2021]; we build on it rather than claim it.

Our starting observation is that this weakness is also an opportunity. The search that exploits the surrogate is the *same machinery* used to generate adversarial examples—a population-based or gradient search that pushes a model's output up. If we can stop it from walking off into unrealistic sequences, the same engine becomes a sequence *designer*. This is the paper's thesis: **attack-as-design**. What turns exploitation into design is not a new search algorithm, but two additions—a rule that keeps the search realistic, and a validation protocol that confirms the designs are real using signals the search never saw.

Existing ways to keep the search realistic add a trained component. One trains a second model to tell training-like sequences from proposed ones, and uses it to lower the score of sequences that look unfamiliar

[Yao et al., 2024]. Another adds a penalty that pushes the surrogate to predict lower scores on the sequences the search proposes [Trabucco et al., 2021]. Both work, and both add an extra piece that must itself be trained or tuned. We ask whether a fixed biological rule, which trains nothing, does as well.

The surrogate itself can be built on a *foundation model*: a network pre-trained on large sequence corpora that maps a sequence to a fixed-length embedding [Lin et al., 2023, Zhou et al., 2024], on top of which a lightweight predictor is trained. We treat this as a swappable choice inside the system, not as the contribution: the featurization can equally be a plain one-hot or k-mer encoding, and we report the comparison as an ablation (Section 3).

Approach. We ask whether a rule that uses no extra trained model is enough. Our rule has two parts, and neither one looks at the surrogate:

1. **Conservative substitutions.** Each substitution relative to the starting sequence must be conservative under the BLOSUM62 (BLOcks SUbstitution Matrix) substitution matrix [Henikoff and Henikoff, 1992], a standard table scoring how often one amino acid replaces another in related natural proteins.

2. **Biophysical window.** Global sequence descriptors—mean hydrophobicity, net charge per residue, aromatic and charged fractions, sequence complexity, and the longest homopolymer run—must lie within the 1st–99th percentile range measured on the training proteins.

A candidate that breaks either part is thrown out during the search. We compare this rule against a hand-tuned penalty (COMs, Conservative Objective Models) and a learned second model (GABO/aSCR, Generative Adversarial Bayesian Optimization with an adaptive Source-Critic Regularization). All methods use the same search algorithm, the same starting points, the same budget, and the same evaluation; they differ only in how they keep the search realistic.

Contributions. (i) **The system works as a designer.** Attack-style search under the biological rule raises the true property on three design tasks—protein thermostability, GFP fluorescence, and 600-base-pair regulatory DNA (all 40 of 40 starting points improved on a true DNA-activity model). (ii) **Every design is independently validated.** On each task the gain is confirmed by a signal the search never used: a held-out predictor ensemble, a separate model family (TemStaPro), a structure predictor (AlphaFold2), kept functional residues, and an exact ground-truth oracle (TF-Bind-8). This surrogate-blind validation protocol is the methodological contribution, not a new optimizer. (iii) **We characterize when the rule helps and when it hurts.** The rule earns its keep only when the search is exploitable and the realistic range is both narrow and aligned with the true objective; a poorly aligned rule can even reduce the true property (we show this on a DNA task). (iv) **The surrogate is swappable.** A frozen foundation-model embedding beats a plain encoding only on long, variable-length protein; on short DNA a one-hot encoding wins. None of the system's claims depend on this choice.

2 Method

Figure 1 summarizes the pipeline. Its components are described below.

Surrogate. The surrogate maps a sequence x to a featurization $\phi(x)$ and then predicts the property with a small neural network m (a multilayer perceptron, MLP). For the protein tasks ϕ is a frozen ESM-2 (Evolutionary Scale Modeling, version 2) [Lin et al., 2023] embedding: we run the foundation model, average its per-residue outputs into one fixed-length vector, and never update the model itself. The featurization is a swappable choice—ϕ can equally be a one-hot or k-mer encoding—and we treat the comparison as an ablation (Section 3), not as part of the system's claims.

Optimizer and objective. We search with a genetic algorithm (GA), a population-based search that maintains a set of candidates, scores them with the surrogate, retains the highest-scoring individuals, and

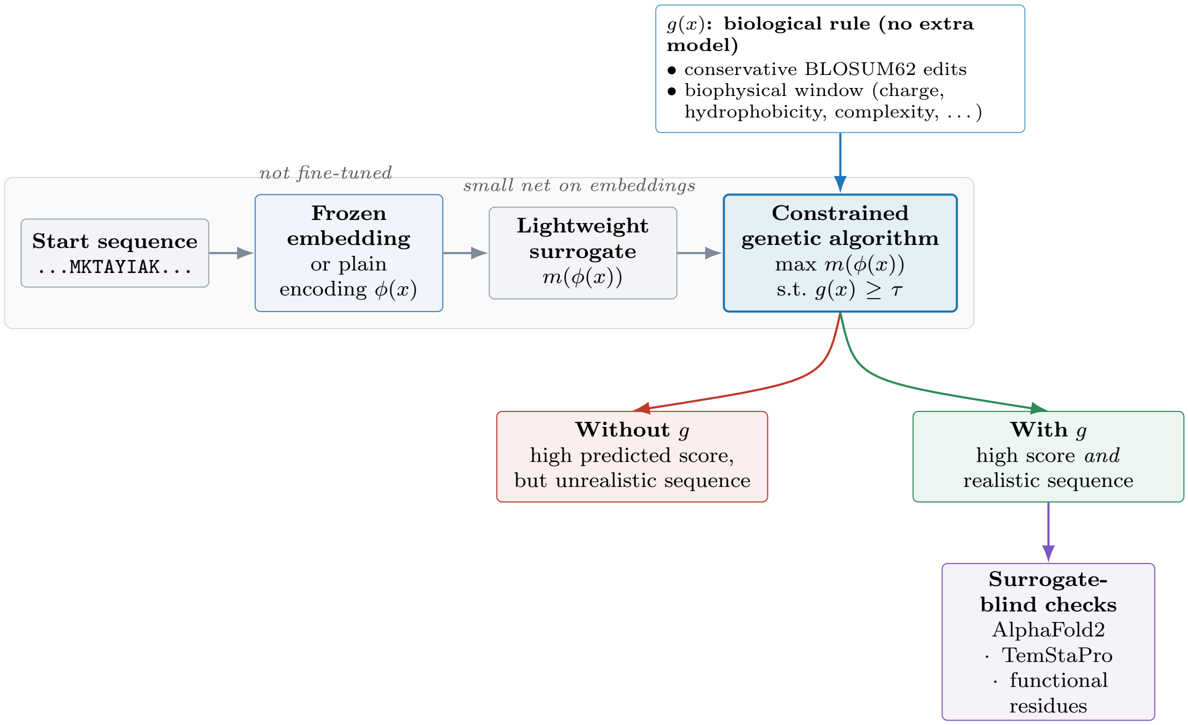

Figure 1: Overview of the attack-as-design system. A starting sequence is mapped to a featurization (a frozen foundation-model embedding, or a plain one-hot/k-mer encoding—a swappable choice); a small surrogate predicts the property; the same search used to generate adversarial examples looks for sequences the surrogate scores highly, subject to a biological rule $g(x) \geq \tau$ (conservative BLOSUM62 substitutions and a biophysical window) that uses no extra model. Without the rule, the search finds sequences that score highly but are unrealistic; with the rule, the sequences it returns are both high-scoring and realistic. Every design is then checked by signals the search never used—the load-bearing step.

generates new candidates from them by changing one residue at a time. Run without any rule, this is exactly the search that tends to find the surrogate's mistakes. We solve

$$\max_x \; m\big(\phi(x)\big) \quad \text{subject to} \quad g(x) \geq \tau,$$

where $g(x) \geq \tau$ is the biological rule. The compared methods differ only in g; the embedding, the search, the starting sequences, the budget, and the evaluation are the same for all of them.

Compared methods. *Unconstrained* just maximizes the surrogate, with no rule. *COMs* [Trabucco et al., 2021] adds a penalty that trains the surrogate to give lower scores to the sequences the search proposes; we tuned its strength for a fair comparison. *GABO/aSCR* [Yao et al., 2024] trains a second model that scores how far a sequence is from the training data, and uses it to discount unfamiliar designs. *Ours* applies the biological rule (conservative BLOSUM62 substitutions and the biophysical window), which trains nothing extra.

Evaluation. We never let the surrogate grade its own designs. We report: (i) *real gain*, measured by a separate set of predictor models that the search never used; (ii) *naturalness* (ΔPLL), the change in the foundation model's masked language-model pseudo-log-likelihood (PLL), a score of how expected a sequence is under the model, where higher values indicate greater naturalness; (iii) *biophysical pass rate*, the fraction of designs within the natural descriptor range; and (iv) for a subset, three fully independent measures: AlphaFold2 [Jumper et al., 2021, Mirdita et al., 2022] predicted local distance difference test (pLDDT), the independent thermostability predictor TemStaPro [Pudžiuvelytė et al., 2024], and preservation of known functional residues.

Table 1: Held-out rank correlation (Spearman ρ) of the property from a frozen foundation-model (FM) embedding versus a trivial encoding, across four tasks.

Task	Input	FM ρ	Simple encoding	Simple ρ	Higher
TF-Bind-8 (SIX6)	8 bp DNA	0.53	one-hot	0.77	simple
TF-Bind-8 (WT1)	8 bp DNA	0.71	one-hot	0.87	simple
Promoter (EPDnew)	300 bp DNA	0.28	4-mer counts	0.29	comparable
Meltome (FLIP)	∼400 aa protein	0.67	amino-acid composition	0.50	FM

3 Experiments

We evaluate the system on four public tasks across two molecule types. Two are protein design tasks. **Meltome** provides melting temperatures for approximately 28,000 proteins from the FLIP (Fitness Landscape Inference for Proteins) benchmark [Dallago et al., 2021, Jarzab et al., 2020]. **GFP** provides fluorescence for approximately 52,000 variants of a single 238-residue green fluorescent protein from ProteinGym [Notin et al., 2023, Sarkisyan et al., 2016]. Their surrogates predict the property from frozen ESM-2 embeddings with rank correlation $\rho \approx 0.67$ (Meltome) and $\rho \approx 0.80$ (GFP). Two are DNA tasks. **Regulatory DNA** designs 600-base-pair sequences for chromatin accessibility, verified on Borzoi [Linder et al., 2025], a published model of DNA activity used here as a true functional oracle. **TF-Bind-8** [Barrera et al., 2016, Trabucco et al., 2022] lists all 65,536 8-letter DNA sequences with their measured transcription-factor binding, giving an exact ground-truth answer for every sequence; we use it as a diagnostic. For each design task we optimize 40 mid-range starting sequences under a shared query budget. The biological rule is adapted to the molecule type: BLOSUM62 substitutions plus the biophysical window for proteins, and a composition band (GC content, repeated-letter cap, and k-mer diversity) derived from the training sequences for DNA.

The core results are the design head-to-heads (Meltome, GFP) and the regulatory-DNA design task, each with independent validation. Two further results scope the system: an ablation on the surrogate featurization, and a diagnostic on the exact TF-Bind-8 oracle. We present the scoping results first because they set up how the design results should be read.

3.1 Ablation: when is a foundation model needed?

The surrogate's featurization is a swappable part of the system. Here we ask when it matters. We compare frozen-embedding surrogates against trivial encodings (a one-hot encoding records each position's letter directly; a k-mer count records how often each short subsequence appears) across four tasks of increasing input length (Table 1, Figure 2). The two short-DNA tasks (TF-Bind-8) measure how strongly a transcription factor binds each 8-base-pair DNA sequence. On 8-base-pair DNA, a one-hot encoding attains higher rank correlation than the DNA foundation model; on 300-base-pair DNA the two are comparable. The foundation model only helps clearly for proteins, whose variable length means a simple encoding cannot be used. This tells us when the method is worth using: on long, variable-length sequences where simple encodings run out.

3.2 A small DNA benchmark, used as a diagnostic

The TF-Bind-8 benchmark [Barrera et al., 2016, Trabucco et al., 2022] lists all 65,536 possible 8-letter DNA sequences together with their measured binding, so the true answer is known for every sequence. On this benchmark every method reaches about the same true score for its best designs, including plain random search (≈ 0.89): the space is simply too small for the methods to pull apart on performance. What they do differ on is how much the surrogate overrates its own designs—the gap between the score the surrogate predicts and the true score (Figure 3). Gradient ascent and AdaLead (a greedy search) overrate the most, while CbAS (Conditioning by Adaptive Sampling) and separable CMA-ES (Covariance Matrix Adaptation Evolution Strategy) stay near zero. We therefore use this benchmark to show the overrating effect, not to rank the methods. This also matches a recent report that the effect does not always appear and has to be checked case by case [Yang et al., 2025].

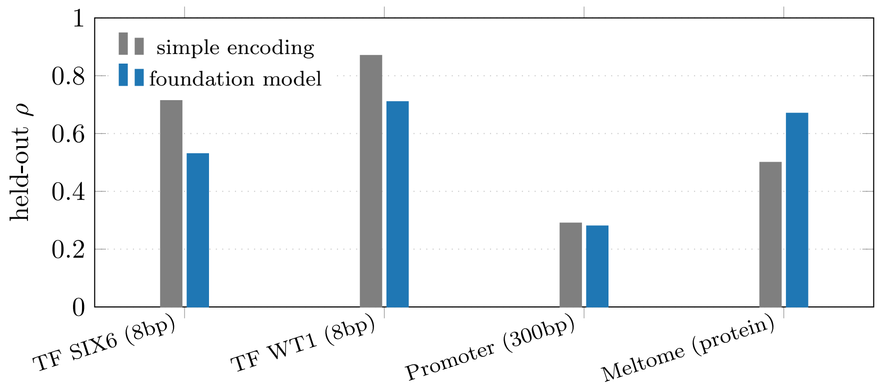

Figure 2: When is a foundation model needed? A surrogate on frozen foundation-model embeddings (blue) beats a simple encoding (grey) only for variable-length proteins. On short DNA the simple encoding is stronger; on 300-bp DNA the two are about the same.

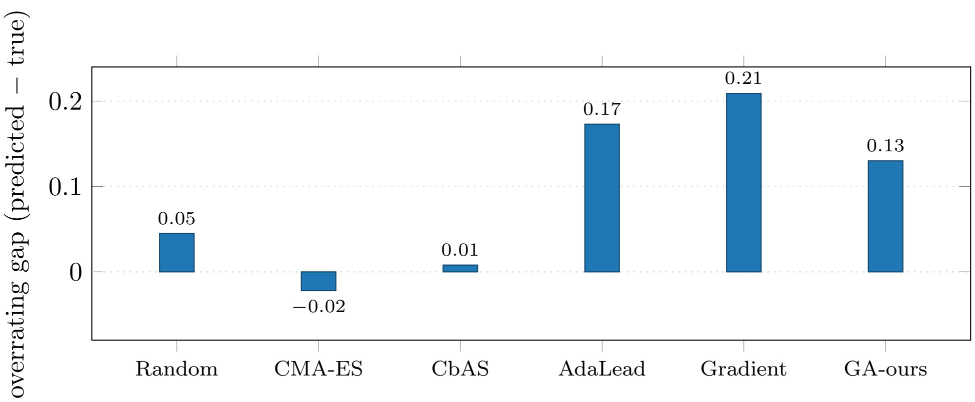

Figure 3: How much each method overrates its designs on TF-Bind-8 (5 seeds, same query budget). All methods reach ≈ 0.89 true score for their best designs, so they do not separate on performance. They differ in how far the surrogate's predicted score sits above the true score; gradient ascent and AdaLead overrate the most.

3.3 The design system works: head-to-head on two protein tasks

Thermostability (Meltome). All four methods reach a similar real gain (+5.9 to +6.8 °C, counting only the starting points that were realistic to begin with). Where they differ is in how natural the designs look (Figure 4, Table 2). Our biological rule gives the most natural designs: it matches the learned second model (GABO; paired Wilcoxon $p = 0.36$, meaning no significant difference) and beats both the hand-tuned penalty (COMs) and the unconstrained search ($p < 0.001$).

Fluorescence (GFP). On GFP the difference is large (Figure 5). Our rule keeps 100% of designs inside the realistic biophysical range, versus 5% (unconstrained), 11% (COMs), and 19% (GABO); all three gaps are significant at $p < 10^{-7}$. The predicted gain is a little lower for our rule (+0.71 versus +0.83 to +0.96)—a small price for designs that are consistently realistic.

3.4 The system generalizes to a third modality: regulatory DNA

To test that the system is not specific to proteins, we applied it to a very different molecule type: designing 600-base-pair regulatory DNA for higher chromatin accessibility. Here the featurization is a frozen Evo2 genomic embedding, the biological rule is the DNA composition band, and the true property is read out by Borzoi [Linder et al., 2025]—a published model of DNA activity that the search never queried, serving as

Table 2: Thermostability head-to-head ($n = 40$ starting sequences; gain is counted over the starting points that were realistic to begin with). The naturalness significance is a paired Wilcoxon test, ours versus each baseline.

Method	Real gain (°C)	Naturalness ΔPLL	Biophysical pass %	vs. ours (naturalness)
Unconstrained	+6.8	−0.191	85	$p < 0.001$
COMs	+6.5	−0.206	88	$p < 0.001$
GABO/aSCR	+5.9	−0.139	85	$p = 0.36$ (n.s.)
Ours	+6.3	**−0.128**	**90**	—

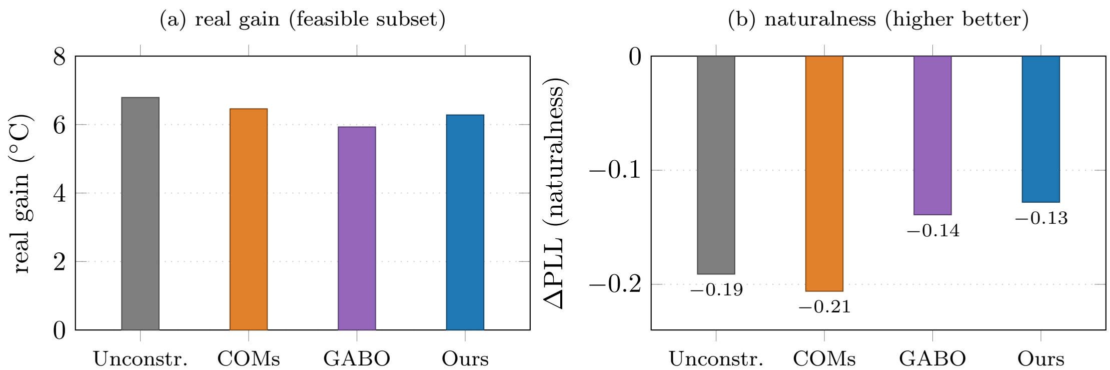

Figure 4: Thermostability head-to-head (Meltome, $n = 40$). (a) Real gains are similar across methods. (b) Our rule gives the most natural designs, matching GABO and beating COMs and the unconstrained search ($p < 0.001$).

the independent oracle. Running the same search loop used for the protein tasks, the system raised the true Borzoi-measured accessibility on **all 40 of 40 starting points** (mean rising from 0.09 to 0.21–0.29 in oracle units). The design engine and the biological rule therefore transfer intact from protein to DNA.

This task also gives a clean second look at the surrogate ablation. At 600 base pairs the genomic foundation model finally decodes the property competitively (held-out $\rho = 0.73$, against 0.70 for a plain k-mer encoding)—a reversal of the short-DNA result in Table 1. Yet in *design* the foundation-model surrogate does not beat the plain encoding: the true gains are statistically indistinguishable (+0.20 for both, unconstrained, paired Wilcoxon $p = 0.48$; +0.12 versus +0.16 constrained, $p = 0.22$). The foundation model's higher *predicted* gain does not convert to a higher *true* gain—the excess is surrogate overrating, exactly the effect the constrained loop and the true oracle are built to catch. This reinforces the ablation's message: the surrogate is a swappable component, and the system's power does not come from it.

3.5 Independent validation

The load-bearing part of the system is that no design claim rests on the surrogate grading its own work: every task is confirmed by a signal the search never used. Across the four tasks these signals are a held-out predictor ensemble, a separate model family (TemStaPro), a structure predictor (AlphaFold2), the known GFP functional residues, an exact ground-truth oracle (TF-Bind-8), and a true DNA-activity model (Borzoi). This protocol—not a new optimizer—is the methodological contribution. The protein-task details follow (Table 3). On thermostability, a separate predictor (TemStaPro) and AlphaFold2 both rank our designs as the most realistic. On GFP, our designs keep all five known important residues intact in 87.5% of cases, versus 67.5–77.5% for the baselines, and keep the key position Ser65 intact every time; the surrogate was never told these positions matter. AlphaFold2 shows that every method still folds into a valid structure (all 10 designs score pLDDT ≥ 70), and our designs change the structure the least (ΔpLDDT −0.6, versus −1.2 to −1.5 for the others).

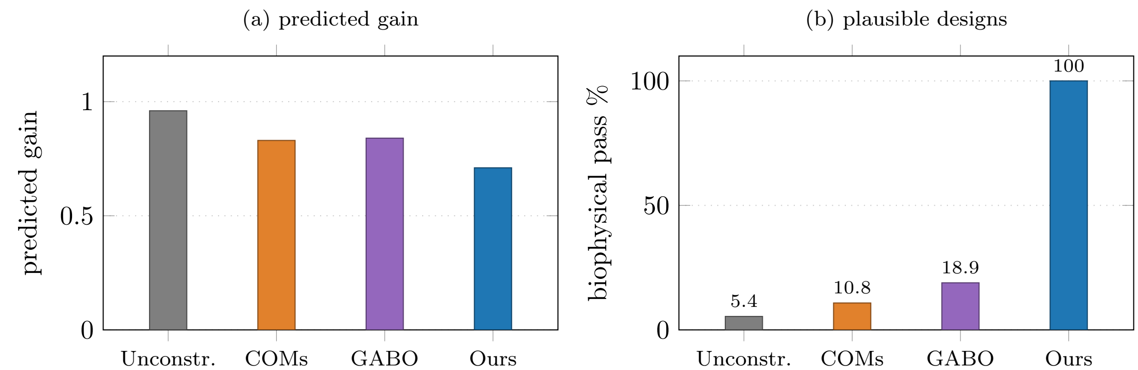

Figure 5: Fluorescence head-to-head (GFP, $n = 37$ realistic starting points). (a) Predicted gains are similar. (b) Fraction of designs inside the realistic biophysical range: 100% for our rule versus 5–19% for the baselines (all $p < 10^{-7}$).

Table 3: Surrogate-blind validation. GFP functional-residue preservation is over $n = 40$ designs; AlphaFold2 ΔpLDDT is over a subset of $n = 10$ (Meltome and GFP each).

Measure	Unconstrained	COMs	GABO/aSCR	Ours
GFP: all 5 functional residues intact (%)	70.0	67.5	77.5	**87.5**
GFP: Ser65 preserved (%)	82.5	87.5	87.5	**100.0**
GFP: AlphaFold2 ΔpLDDT vs. start	-1.45	-1.26	-1.22	$\mathbf{-0.64}$

3.6 When the rule helps, and when it hurts

The constraint improves naturalness on Meltome and the biophysical pass rate on GFP. This difference has a simple cause: how narrow the range of realistic sequences is (Figure 6). GFP is a single 238-residue protein, so its biophysical range is 8–19 times narrower than Meltome's, which spans thousands of distinct proteins. When the range is narrow, random changes leave it quickly (after 26 edits, only 3% of GFP designs are still in range, versus 99% for Meltome), so the biophysical check is the one doing the work. When the range is wide, most candidates pass it, so the biophysical check no longer separates methods and the naturalness of the design does instead. In short, the constraint helps through whichever check is hard to pass, and which check that is depends on how many distinct proteins the training data covers: one protein gives a narrow range, thousands give a wide one.

When the rule hurts: the range must match the true objective. The rule is not free. It only helps when the realistic range it enforces actually contains the genuinely good sequences—when the range is *aligned* with the true objective. If the range excludes good sequences, the rule raises the pass rate while lowering the true property: it looks like compliance but is really the constraint fighting the objective. We see this on a DNA binding task (WT1), whose true binding motif is a run of cytosines. The generic composition band caps repeated letters, so it rejects most of the strongest true binders; the rule then reports 100% "realistic" designs while the true binding drops. The practical lesson, which we followed for the regulatory-DNA task above, is to derive the range from the training distribution of the actual task rather than from a generic rule, so that the range and the objective agree.

4 Related work

Designing sequences by maximizing a learned surrogate, and the tendency of that search to find the surrogate's mistakes, are both well studied [Trabucco et al., 2021, 2022, Fu and Levine, 2021]. Known fixes include the hand-tuned penalty [Trabucco et al., 2021], a learned second model that scores how training-like a sequence is [Yao et al., 2024], and biological priors built into the way sequences are sampled [Kmicikiewicz

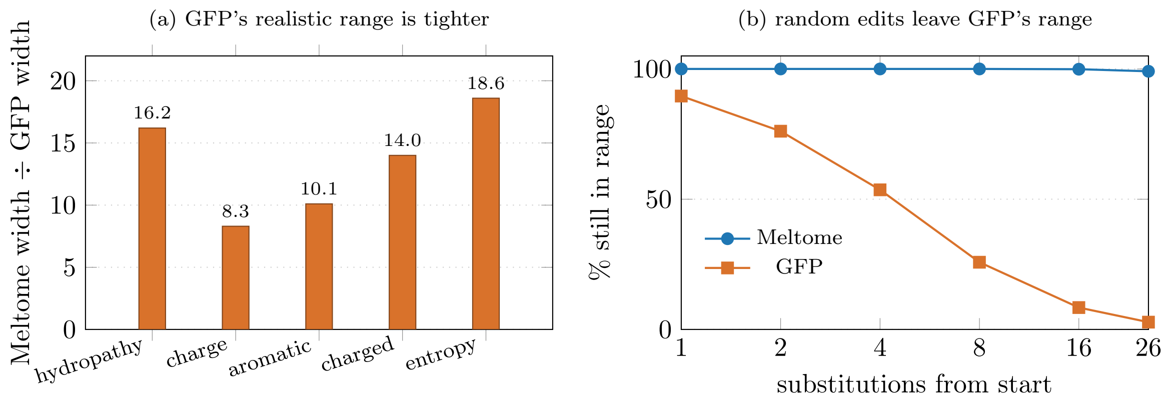

Figure 6: Why the constraint helps through a different check on each task. (a) GFP's range of realistic sequences is 8–19× narrower than Meltome's on every biophysical descriptor, because GFP is one protein and Meltome is thousands. (b) Random substitutions leave GFP's narrow range quickly but stay inside Meltome's wide one, so the biophysical check is the hard one for GFP and naturalness is the hard one for Meltome.

et al., 2025]. Recent work measures how these surrogates fail on biological sequences [Surana et al., 2024] and reports that the failure does not happen in every setting [Yang et al., 2025]. Foundation-model sequence design is also advancing for DNA and RNA [Zhang et al., 2026, Wang et al., 2024]; those methods keep training the language model further and search in a learned internal space, whereas we keep the model frozen and search directly over sequences. Finally, genomic foundation models are known to be foolable when used as classifiers [Yoo et al., 2024, Luo et al., 2025, Zhan et al., 2025, Krishnan et al., 2026]—the same weakness our search exploits and our validation guards against. In our own earlier work we used exactly this attack—a black-box genetic search over sequences—to show that adversarial genomic sequences can evade biosecurity screening [Krishnan et al., 2026]; here we turn the same engine around and ask what it takes to make it *design* rather than merely fool. Our contribution is not a new optimizer but a *system*: reusing attack-style search as a design engine, a training-free biological rule that keeps it realistic as well as a learned second model does, a surrogate-blind validation protocol demonstrated across four tasks and two molecule types, and a characterization of when the rule helps and when a misaligned rule hurts.

5 Limitations

All our tests are computational; testing designs in the lab is future work, and our independent checks are themselves models, though the TF-Bind-8 oracle is an exact ground-truth lookup. The biophysical check looks at whole-sequence numbers and may miss structural details, which is partly why we also run AlphaFold2 and the important-residue check. Our most expensive checks use small samples (n = 10–16), though the main design and plausibility results use all 40 starting points per task. The search machinery is assembled from known pieces; the contribution is the validated, characterized system, not a new optimizer. Finally, as the mechanism section shows, the rule helps only when its range is tight and aligned with the true objective—a misaligned range can be cosmetic or even counterproductive, so the range must be built from the task's own training distribution.

6 Conclusion

The search that produces adversarial examples can be turned into a biological sequence designer by adding two things: a training-free biological rule that keeps the search realistic, and a validation protocol that confirms every design against signals the search never used. The rule matches a learned second model and beats a hand-tuned penalty, at a similar predicted gain and without training anything extra. The system raises the true property on three design tasks spanning proteins and DNA—confirmed on every task by an independent signal, including an exact oracle and a true DNA-activity model—and we characterize when the rule helps and when a misaligned rule hurts. Which featurization the surrogate uses is a swappable component that changes none of these claims.

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